



Mini_Project2

Inventory Management System

Project Description

Mini-project 2 involves building an **Inventory Management System** using **.NET Core** with **Entity Framework Core**. The system will manage products, categories, and suppliers, supporting:

- **CRUD operations** for each entity
- **Filtering, sorting, and pagination**
- **Repository pattern** with **Unit of Work** for clean data access
- **Data validation** (FluentValidation or Data Annotations)

Core Entities (Tables)

1. Product

- **Fields:**
 - Id (int, PK)
 - Name (string, required)
 - Description (string, optional)
 - Price (decimal, must be > 0)
 - StockQuantity (int, must be ≥ 0)
 - CategoryId (int, FK to Category)
 - SupplierId (int, FK to Supplier)

2. Category

- **Fields:**
 - Id (int, PK)
 - Name (string, required, unique)
 - Description (string, optional)

3. Supplier

- **Fields:**
 - Id (int, PK)
 - Name (string, required)
 - ContactEmail (string, must be valid email)
 - Phone (string, optional)

Features to Implement

1. Repository Pattern

- **Repositories:**
 - IProductRepository (with CRUD + filtering)
 - ICategoryRepository (basic CRUD)
 - ISupplierRepository (basic CRUD)
- **Unit of Work (IUnitOfWork)**
 - Manages transactions across repositories
 - Ensures single database context per request

2. Filtering & Pagination

- **Product Filtering Options:**
 - By Name (partial match)
 - By CategoryId
 - By SupplierId
 - By Price range (min/max)
 - By Stock availability (in-stock/out-of-stock)
- **Pagination:**
 - Default page size = 10
 - Page number starts at 1

3. Sorting

- **Sort Products By:**
 - Name (A-Z / Z-A)
 - Price (low-high / high-low)
 - StockQuantity (asc/desc)

4. Data Validation

- **FluentValidation** (or Data Annotations) for:
 - Product: Price > 0, StockQuantity ≥ 0
 - Category: Name is unique
 - Supplier: ContactEmail is valid

Submission Guidelines

- Submit:
 - Source Code (.zip)
 - Demo Video (2-3mins, Show the core features of the project)

Thank You
Edges For Training Team